George Steuart Health — Dashboard

Complete Server Deployment & Operations Guide

A comprehensive, production-tested manual covering architecture, initial deployment, background services, reverse proxying, credential governance, and 1-click update pipelines.

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1. Server & System Architecture Overview

Component	Specification / Configuration
Server Host IP	`172.16.7.10`
Domain / URL	`http://gsh-sd.georgesteuart.lk` (Local Corporate Network)
SSH User	`tmsadmin`
Project Root Directory	`/var/www/dashboard`
Backend Directory	`/var/www/dashboard/backend`
Frontend Directory	`/var/www/dashboard/frontend`
Backend Service	`dashboard.service` (Systemd / Uvicorn ASGI on Port 8000)
Web Server / Proxy	Nginx (Port 80) serving `/frontend/dist` & proxying `/api`
Database	MySQL (Database: `gsh_dashboard` on Port 3306)
External Integrations	Oracle ERP Database (Live Invoices & Backlog), Axienta SFA Cloud

Default System Login Credentials

Role	Username	Email	Default Password	Permissions / Access Level
■■ **Admin**	`admin`	`admin@gsh.lk`	`Admin@GSH2026!`	Full administrative privileges (Formula guides, user management, sync, upload, mappings)
■ **Standard**	`user`	`user@gsh.lk`	`User@GSH2026!`	Dashboard executive viewer access (View charts, division reports, backlog filters)
■ **SSO**	Corporate	`name@georgesteuart.lk`	Office 365 Pass	Microsoft Entra ID (Azure AD) Single Sign-On Authentication

2. Initial Server Setup & Deployment Steps

Step 1: Install Server Prerequisites & System Packages

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y python3 python3-pip python3-venv nodejs npm nginx mysql-server git libaio1
libaio-dev curl
```

Step 2: Install Oracle Instant Client (For Oracle DB Integration)

Oracle Client drivers are required to connect directly to the Oracle ERP database for live Invoices and Backlog data retrieval:

```
# 1. Create directory and extract Oracle Instant Client files
sudo mkdir -p /opt/oracle
cd /opt/oracle
# (Extract instantclient-basic-linux.x64-19.x.zip into /opt/oracle)

# 2. Configure Environment Variables & Library path
echo "/opt/oracle/instantclient_19_24" | sudo tee /etc/ld.so.conf.d/oracle-instantclient.conf
sudo ldconfig

# Symlink fix for libaio on Ubuntu 24.04+:
sudo ln -s /usr/lib/x86_64-linux-gnu/libaio.so.1t64 /usr/lib/x86_64-linux-gnu/libaio.so.1 || true
sudo ldconfig
```

Step 3: Setup Project Directory & Clone Codebase

```
sudo mkdir -p /var/www/dashboard
sudo chown -R $USER:$USER /var/www/dashboard
cd /var/www/dashboard
# (Copy or clone codebase into /var/www/dashboard)
```

Step 4: Backend Virtual Environment & Dependencies Setup

```
cd /var/www/dashboard/backend
python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

Backend Environment File (/var/www/dashboard/backend/.env):

```
DB_HOST=127.0.0.1
DB_PORT=3306
DB_USER=root
DB_PASSWORD=your_mysql_password
DB_NAME=gsh_dashboard
ORACLE_USER=YOUR_ORACLE_USER
ORACLE_PASSWORD=YOUR_ORACLE_PASSWORD
ORACLE_DSN=172.16.x.x:1521/ORCL
AXIENTA_API_URL=https://api.axienta.lk
```

Step 5: Configure Systemd Background Service

Create /etc/systemd/system/dashboard.service to keep Uvicorn running 24/7:

```
[Unit]
Description=GSH Dashboard FastAPI Backend Server
After=network.target mysql.service

[Service]
User=root
WorkingDirectory=/var/www/dashboard/backend
ExecStart=/var/www/dashboard/backend/.venv/bin/uvicorn app.main:app --host 127.0.0.1 --port 8000
--workers 4
Restart=always
RestartSec=5
EnvironmentFile=/var/www/dashboard/backend/.env

[Install]
WantedBy=multi-user.target
```

Enable and Start the Service:

```
sudo systemctl daemon-reload
sudo systemctl enable dashboard
sudo systemctl restart dashboard
sudo systemctl status dashboard
```

Step 6: Frontend Build & Nginx Configuration

Frontend Production Environment (/var/www/dashboard/frontend/.env.production): VITE_API_URL=/api

```
cd /var/www/dashboard/frontend
npm install
npm run build
```

Nginx Site Configuration (/etc/nginx/sites-available/dashboard):

```
server {
listen 80;
server_name gsh-sd.georgesteuart.lk 172.16.7.10;

root /var/www/dashboard/frontend/dist;
index index.html;

location / {
try_files $uri $uri/ /index.html;
}

location /api/ {
proxy_pass http://127.0.0.1:8000/api/;
proxy_http_version 1.1;
proxy_set_header Upgrade $http_upgrade;
proxy_set_header Connection 'upgrade';
proxy_set_header Host $host;
proxy_cache_bypass $http_upgrade;
proxy_set_header X-Real-IP $remote_addr;
proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
proxy_read_timeout 300s;
proxy_connect_timeout 300s;
}
}
```

Activate Site and Reload Nginx:

```
sudo ln -sf /etc/nginx/sites-available/dashboard /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default
sudo nginx -t
sudo systemctl restart nginx
```

3. How to Deploy New Updates to the Server

Method 1: Manual Step-by-Step Deployment

```
# 1. Connect via SSH
ssh tmsadmin@172.16.7.10

# 2. Pull Latest Code
cd /var/www/dashboard
git pull origin main

# 3. Update Backend Dependencies and Restart
cd /var/www/dashboard/backend
source .venv/bin/activate
pip install -r requirements.txt
sudo systemctl restart dashboard

# 4. Rebuild Frontend Assets and Reload Nginx
cd /var/www/dashboard/frontend
npm install
npm run build
sudo systemctl reload nginx

# 5. Verify Health Endpoint
curl -s http://127.0.0.1:8000/api/health
```

Method 2: Automated 1-Click Update Script (Recommended)

An automated shell script /var/www/dashboard/deploy.sh is available on the server to execute all update and verification steps in one sequence:

```
#!/bin/bash
set -e
echo "============================================="
echo "■ Starting GSH Dashboard Production Deploy"
echo "============================================="
cd /var/www/dashboard

# 1. Pull latest code
echo "■ 1. Pulling latest git updates..."
git pull origin main || echo "■■ Git pull skipped or not configured."

# 2. Update Backend
echo "■■ 2. Updating Backend..."
cd /var/www/dashboard/backend
source .venv/bin/activate
pip install -r requirements.txt --quiet
sudo systemctl restart dashboard

# 3. Update Frontend
echo "■ 3. Building Frontend Assets..."
cd /var/www/dashboard/frontend
npm install --silent
npm run build
sudo systemctl reload nginx

# 4. Verification
echo "■ 4. Verifying Health Status..."
sleep 2
HEALTH=$(curl -s http://127.0.0.1:8000/api/health)
echo "Backend Status: $HEALTH"
echo "============================================="
echo "■ Deployment Completed Successfully!"
echo "■ Dashboard URL: http://gsh-sd.georgesteuart.lk"
```

```
echo "========================================="
```

To execute any future update, run simply:

```
sudo /var/www/dashboard/deploy.sh
```

4. Common Troubleshooting & Useful Commands

Action / Task	Command
View Live Backend Logs	`sudo journalctl -u dashboard -f`
View Nginx Error Logs	`sudo tail -f /var/log/nginx/error.log`
Restart Backend Service	`sudo systemctl restart dashboard`
Reload Nginx Server	`sudo systemctl reload nginx`
Check MySQL Service Status	`sudo systemctl status mysql`
Check Port 8000 Activity	`sudo netstat -tulpn | grep 8000`
Check Oracle Client Libs	`ldconfig -p | grep libclntsh`
Test Backend Health Endpoint	`curl -s http://127.0.0.1:8000/api/health`